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Comparative Analysis of Petroleum Fiscal System in Africa (A Case Study of Nigeria, Angola, Algeria, and Libya)

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ABSTRACT

International Oil Companies (IOCs) are constantly interested in expanding or diversifying their investment portfolios and as such searching for markets that are favourable for these purposes. A major determinant in the making of investment decisions in the oil and gas industry in any country or region is the Petroleum Fiscal System (PFS). It helps describe the relationship between the Host Government (HG) of the oil-producing region and the IOC involved in terms of profits sharing. For this purpose, finding the comparative analysis of each country's fiscal system for performance becomes imperative as it directly involves the IOCs. In this study, a comparative performance analysis of fiscal systems is carried out for the four largest producers of Oil in Africa namely; Nigeria, Angola, Algeria, and Libya. An economic model is developed for these countries for a 25 years' timeframe incorporating features of each fiscal system. A wide range of economic indicators were used for the purpose of comparison in this study such as Internal Rate of Return (IRR), Net Present Value (NPV), Discounted Net Cash Flow (DNCF), Profitability Index (PI), Contractor take (CTake), Government Take (GTake) and Payback Period (PP). A brief summary of the results gotten show that Algeria had the most NPV of the four countries with $8.3 billion and Nigeria had the lowest with $2.3 billion. Angola had highest Contractor take (CTake) to be 27.21%. Generally, all four countries had positive returns on investments with each having different rates. This gives HG, Petroleum analysts and most importantly IOCs a varied option of performance indicators for making policy decisions. Also a SWOT analysis is carried out in terms of their management structure, geographical location, political and social state of each country.

Keywords: Petroleum Fiscal System, Africa, Nigeria, Angola, Algeria, Libya.

Introduction

The estimated maximum capacity for energy production given all available resources on earth are what is refer to as world energy resources. They can be classified by type into fossil fuel, nuclear fuel and renewable resources. Enough statistical evidence abounds to show that the continent Africa is blessed with vast quantities of fossil resources. Africa has the richest concentration of natural resources such as oil, copper, diamonds, bauxite, lithium, gold, hardwood forests, and tropical fruits. It is estimated that 30% of the earth's mineral resources are found in the African continent and it is estimated that 12% of the oil produced in the world comes from Africa; (AFDB 2015). The countries that lead in oil production in Africa are Nigeria, Angola, Algeria, Libya and Egypt and these nations have accounted for more than 8.7 million barrels of all the oil produced daily in 2014. Nigeria and Libya are ranked among the top ten countries with the largest oil reserves in the world at 48 billion barrels for Libya and 37 billion barrels for Nigeria.

The need for global and rapid urbanization and industrialization, quest for economic growth and the attainment of energy security, independency and sustainability has led to an increase in the exploration and transportation of crude oil. Weber and Jeremy (2012) argued that for a country or region to be thought of as been industrialized, energy must have been used for the production of goods and services, and to generate income and employment. This shows the importance of oil exploration considering the fact that most material used as finished products in almost all industries have their sources of raw materials gotten from products of oil exploration. Due to this increase in oil and gas exploration over the last decade, complexity surrounding the policies and fiscal systems behind its exploration and production has been on the increase. World Bank experts have argued that the presence of hydrocarbon in commercial quantities affects a country politically, socially, economically and strategically (Bastida, 2005).

To sustain the development of oil and gas resources requires policies, principles and practices that will support the utilization of these resources in a way that will benefit both this present generation and the future. These policies and principles are what are referred to as fiscal systems. Fiscal systems provide terms and conditions of conducting business worldwide during the complete integrated process from exploration to production and trading. Fiscal terms are the most important terms of a natural resource contract as they delimit and define the amounts of profit and economic rent that will accrue to each party throughout the life of the contract. The ownership of mineral rights in most African countries is entirely vested in the National or host government. Based on this, federal petroleum laws form the backbone for all petroleum related operations (Echendu, 2012). Generally, Oil Ministries or National Oil companies (NOC) represent the host government (HG). These entities are vested with the power to grant licenses or enter into contractual agreements with International oil companies (Onwuka, 2012). The main objective of an IOC is to ensure consistent return on investments equal to the risk associated with projects undertaken and their organization's strategic objectives, while host government aim to achieve national development and socioeconomic objectives while maximizing rent for their country through efficient fiscal systems.

These fiscal systems can be grouped under two families: concessionary systems and contractual systems. In both systems, the investor assumes all risks and costs associated with hydrocarbon exploration, development and production, and receive compensation adequate to the risk. Normally, the investment risks are assumed by oil companies rather than the state/owner of the resource. In general terms, the higher the risk of investment activities in a country, the higher the portion of the rent received by the investor. The fundamental difference between concessionary and contractual systems relates to the ownership of the natural resources.

Under a concessionary system, the title to hydrocarbons passes to the investor at the borehole. The state receives royalties and taxes in compensation for the use of the resource by the investor. Title to and ownership of equipment and installation permanently affixed to the ground and/or destined for exploration and production of hydrocarbons generally passes to the state at the expiry, or termination, of the concession (whichever is earlier). The investor is typically responsible for abandonment.

Under a contractual system, the investor acquires the ownership of its share of production only at the delivery point. Title to and ownership of equipment and installation permanently affixed to the ground and/or destined for exploration and production of hydrocarbons generally passes to the state immediately. Furthermore, unless specific provisions have been included in the contract the government (or the national oil company, "NOC") is typically legally responsible for abandonment.

Aim of the Study

This study is aimed at analyzing the relative attractiveness of Oil exploration and production in Africa with particular reference to Nigeria, Angola, Algeria and Libya as the biggest producers of crude oil in Africa as a whole. The following were done to achieve the stated aim:

- To appraise a comprehensive review on petroleum fiscal systems and structures globally contributing to the global debate on fiscal systems.
- To carry out a SWOT analysis.
- To critically compare fiscal systems in Nigeria, Angola, Algeria and Libya in terms of economic tools.
- To make recommendations by which prospective or already existing foreign investors can make justified explorations and production decision in these four regions of Africa.

In order to effectively carry out this study, this work is focused on developing fiscal models using MS-EXCEL. The model developed will integrate the elements of the fiscal provisions of the various fiscal systems.

Literature Review

What is Fiscal System?

The petroleum fiscal regime of a country is a set of laws, regulations and agreements which governs the economic benefits derived from petroleum exploration

and production. The regime regulates transactions between the political entity and the legal entities involved (Johnson, 1994). A commercial or legal entity in this context is commonly an oil company, and two or more companies may establish partnerships to share economic risks and investment capital. Although petroleum, oil and gas, and hydrocarbons are not technically mineral resources, the term mineral rights are used to denote rights to exploit oil and gas resources from the underground (Mian, 2010). Onshore, in United States, the landowner possesses exclusive rights for mineral rights, elsewhere generally the state does. The petroleum licensing system of a country may be considered interwoven with the fiscal regime; however, a licensing system has its distinct function: to grant rights for petroleum exploration and production to commercial entities. Because each country has distinctive legislation, there are theoretically just as many different fiscal regimes as there are countries in the world with petroleum resources, but the regimes can still be categorized based on their common characteristics (Carlos, 2005).

Classification of Petroleum Fiscal Systems

Petroleum fiscal systems whereby the owner of mineral resources receives levies from the extraction company can be classified into two main categories These are concessionary systems and contractual systems as shown in Figure 1.

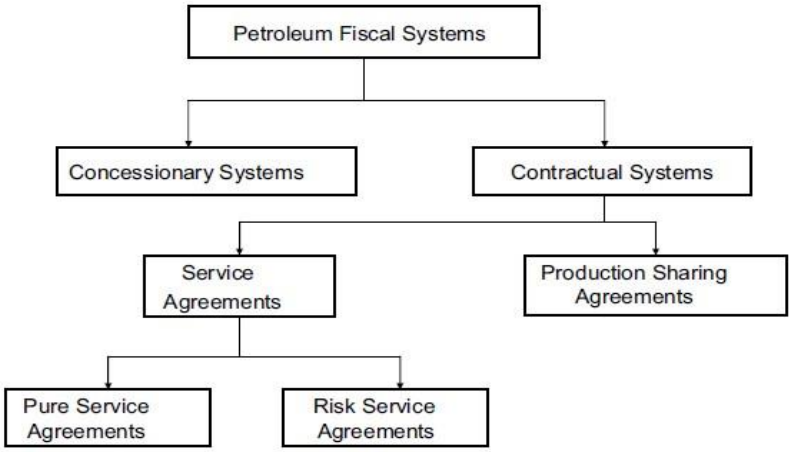

Figure 1: Classification of fiscal system

In most countries, except the United States of America, the owner of the mineral resources is the government. In the USA, the owners are private individuals or companies that pay taxes on production to the state. Worldwide, every country has developed its own petroleum fiscal systems to be applied.

Concessionary System

Under concessionary systems, the government will transfer title of the oil and gas to a company if they are produced. The producing company then pays royalties and taxes. Also the state government grants a Concession or License to an international oil company

(IOC) or a consortium which gives rights for a fixed period to explore and produce hydrocarbons within a certain area (License Area or Block). The IOC may be required to pay a signature bonus or a license fee to the government to secure the Concession or License. Thereafter, the government will obtain compensation usually through royalty and tax payments when hydrocarbons are produced (Yinka,2005).Concessionary systems are used by around half of the countries worldwide including the US, UK, France, Norway, Russia, Australia, New Zealand, South Africa, Colombia, and Argentina. These countries have fiscal regimes which vary widely in terms of royalty and tax rates, tiers of taxation and other features such as incentives to promote investment.

Contractual System

Contractual systems are in most cases either:

a) Production sharing agreements
b) Service contracts

The private companies under contractual systems have the right to receive a share of production or revenues from the sale of oil and gas in accordance with a production sharing agreement (PSA) or a service agreement (SA). The state companies either self-produce or share the production and selling of the oil or gas. Revenues then flow into the finance ministries' treasuries (Yinka, 2005). In most contractual systems, the facilities installed by the contractor within the host government's territory become the property of the state either as soon as they are landed or upon start up or commissioning. Sometimes the asset or a facility does not pass to the government until the expended costs have been recovered. This transfer of title for asset facilities does not apply to leased equipment or to equipment brought in by service companies.

The difference between service contracts and production sharing contracts depends on whether the contractor receives compensation in cash or in crude. Under a **Production Sharing Agreement,** the contractor receives a share of production and hence takes title to this crude. In a service contract, there is no issue of title since the contractor gets a share of profits rather than production. Under some **Service Agreements,** however, the contractor has the right to purchase crude from the government at a discount. Despite the differences between the systems, the same economic results are achieved (Johnston, 2003).

Provision of an Oil and Gas Fiscal Regime

These are the provision in a typical oil and gas fiscal regime, they include;

1. The size, the shape and the geological limit of the acreage to be acquired, explored and developed.
2. The kind of contract (concessionary or contractual contract).
3. The kind of permit (Oil Prospecting License (OPL) for 5 years and Oil Mining License (OML) for 25-30 years), in some cases, an OPL is given and when oil is found in commercial quantity, it is upgraded to OML.
4. Fees and bonus (surface fees, rentals, production and signature bonus).
5. The Obligations.

6. Employment and training of locals.
7. Assignment or transfer of acreage.
8. Royalty agreement, profit sharing and cost recovery/cost deduction.
9. Obligations to supply domestic market and build local refineries.
10. Equity participation.
11. Initial, primary and extension of field and acreages.
12. Drilling relinquishment, surrender and abandonment.

Elements of Fiscal Regimes

For most countries, a selection of the elements of fees and taxes listed below applies, very few countries, if any, have implemented all elements.

- **Bonuses:** There are different flavors: signature bonus, discovery bonus, first oil sales, and production bonus. Signature bonus is a onetime fee for the assignment and securing of a license, paid irrespective of economic success for the contractor or licensee. Not all states use bonuses, but the government may charge a minor fee for handling license applications.
- **Corporate tax:** Corporate tax is the standard company income tax used in many countries, and will similarly apply to oil companies.
- **Royalties:** Royalties are shares of the extracted hydrocarbons entitled to the host state. The state can agree with the licensees to take it in kind or in cash. This arrangement applies to both crude oil and to natural gas, both in concessionary and contractual license systems.
- **Production shares:** The body of a production sharing contract layouts the production share between the contractor(s) and the state or its state-owned oil company. Typically, most of the early production will be set aside for recovering the costs incurred during development by the contractor (cost oil), while the state receives an increasing share of production after costs are recovered (profit oil). This is a specific contractual license system arrangement.
- **Surface fee:** Surface fee is a yearly fee, paid per square kilometer or square mile occupied by the license or leased area. This type of fee is used in Brazil for the exploration phase, and for large production volumes, named “Occupation or Retention Fees". In the Norwegian fiscal regime, it is known as area rental fee and is only paid for "passive licenses" and for exploration area before a plan for development and Operations is submitted to the government. If an oil company has found all or parts of an exploration area of little interest, the area can be relinquished to the state to save expenses for fees. Other countries enjoying surface fee include Algeria, Angola, Benin, Cameroon and Mauritania.

Government/Contractor Take

A very important variable in negotiating a contractual agreement between an IOC and the Host Government is determining how much of the project’s revenue will be allocated to each party over the lifetime of the project. Kaiser and Pulsiphur (2004) define the division of a projects profit between Host Governments and IOCs as “Take”. For the take of a particular project to be calculated, it is imperative to be able to forecast

the expected cash flow for the project. This has proved to be rather difficult to achieve as getting an estimate cash flow for a proposed project is highly prone to errors due to incomplete information concerning the field to be explored. The "Take" is calculated based on economic, geographical or political conditions existing at that particular point in time and as such be renegotiated as these parameters change or more information is gotten. A 50% - 50% split between Host Governments and IOCs was considered to be a fair bargain before the two oil shocks as claimed by Rutledge and Wright (1998), the creation of the Organisation of Petroleum Exporting Countries (OPEC) has led IOCs to accepting a lesser percentage of take. Host Governments are more concerned with "take" as it is more of a fiscal statistics and not an economic statistic (Kaiser and Pulsiphur 2004). Due to the fact that "take" is not a direct indicator of the economic performance of a project, IOCs take it as a secondary interest (Johnston 1994a). Technically, the riskier the project, the greater the share of profit demanded by the IOCs.

Profitability Analysis in Oil and Gas Investment (Capital Budgeting Techniques)

Capital refers to expenditure disbursed at the beginning of an investment. It refers to the cash amount used in the purchasing of physical assets like plants, equipment, properties, acquisition of license etc. Budgeting refers to the plan that gives detail of projected cash inflows resulting from projected cash outflows. Hence "Capital budgeting" is the deliberate allocation of present resources with an expectation of futures returns (Mian, 2002).

In oil and gas industries, projects are capital intensive and involve disbursement of huge funds therefore proper budgeting and planning is necessary. To achieve this, there are lain down rules and techniques to follow, these techniques are referred to as profitability analysis. In the oil and gas industry capital expenditure (CAPEX) includes the cost of exploration, new oil and gas field development, infilling drilling, facilities replacement, upgrade of facilities, installation of new facilities such as Artificial lift systems or pressure maintenance systems, secondary or enhanced oil recovery and a variety of downstream components, gas treatment plants and separation components.

Capital budgeting techniques involves estimating the net worth of future cash flows and other benefits as compared to the initial investments. The tools used for these evaluations are referred to as Profitability tools or economics tools. There are quite a large number of economic evaluation tools, they have different properties and are used in different ways.

They include:

1. Net Present Value (NPV)
2. Internal rate of return (IRR)
3. Profitability Index (PI)
4. Present value ratio (PVR)
5. Payback Period

Net Present Value (NPV)

Net Present Value (NPV) refers to as the present value of a cash flow. It is obtained by subtracting the present value of periodic cash outflow from the present value of alternative use of funds. By alternative use of funds, we mean the value loss with time because of the missed opportunity of any interest. If the NPV of an investment at a

certain discount rate is positive it means that it pays the cost of financing an investment or the cost of alternative use of funds. However, if the NPV is negative, it means that the investment is not generating funds equivalent to those expecting from the alternative use of funds, thus alternative loss occurs. The calculation of NPV is made easy with the demonstration of net cash flow.

NPV can be calculated using the equation;

$$NPV = \frac{NCF}{(1+i_e)^t} \quad (1)$$

Where: NCF= Net Cash Flow

i_e- Discount rate %

t - Time (t in years, month)

Once the NPV of an alternative investment is gotten, the decision maker applies certain rules, which are:

- If the NPV is positive, accept proposal.
- If the NPV is negative, reject proposal.
- If the NPV is zero, the decision may be indifferent because at this point, the investment is making neither profit nor loss, but it is self-sustaining.

Internal Rate of Return (IRR)

This is a profitability indicator widely used to measure the profitability of project and business venture. It is also refer to as Rate of Return (ROR) or Internal Yield or Marginal Efficiency of Capital. Internal rate of return is the discount rate at which the net present value is exactly equal to zero or it is the discount rate at which the present value of cash inflow is equal to the present value of cash outflow.

In calculating IRR, a trial and error method is used because varying discount rate is applied on the same project to check which discount rate would produce an optimum balancing position. It can also be calculated by plotting series of net present values (cash flows) vs. discount rate.

There are two ways to calculate the IRR, they include:

1. Graphical Method
2. Trial and Error Method

The trial and error formula is stated below:

$$IRR = r_a + \frac{NPV_a}{(NPV_a - NPV_b)}(r_b - r_a) \quad (2)$$

Where: IRR = Internal Rate of Return

r_a = lower discount rate chosen

r_b = higher discount rate chosen

NPV_b = NPV at r_b

NPV_a= NPV at r_a

Criteria for using IRR in investment decision making;

If the investment is financed by a hundred percent (100%) borrowed capital the IRR should be greater than or equal to the interest rate charged by the bank for borrowing that fund.

Summarily;

- IRR > interest rate accepts the investment.
- IRR < interest rate rejects the investment.

Profitability Index (PI)

Profitability index is a dimensionless ratio obtained by dividing the present value of the future operating cash flow by the net present value of the investment. It can be calculated by using the expression below:

$$\text{PI} = 1 + \frac{\text{NPV}}{\text{PV of capital investment (\$)}} \quad (3)$$

Present Value of capital investment (PV CAPEX) is the discounted capital expenditure invested in the business.

PI shows the size of the initial investment and the efficiency of the investment. It answers the question of how much revenue is generated per dollar or naira invested. PI shows which investment is more profitable assuming two positive NPVs. PI shows the relative profitability of the investment.

The criteria for Using PI in Investment analysis are as follows:

- Accept all investment proposals with PI >1
- Reject all investment proposals with PI < 1
- If PI = 1, the projects is self-sustaining, i.e. it makes neither profit nor loss
- If all projects have PI > 1, accept the investment with the highest PI.
- PI is used for capital allocation because the project with the highest, PI is most profitable.

Present Value Ratio (PVR)

This is the ratio of the NPV to the present value of the capital investment. It can be calculated as follows:

$$\text{PVR} = \frac{\text{NPV}}{\text{PV of capital investment (\$)}} \quad (4)$$

Criteria for applying PVR in investment analysis:

- For all project with PVR > 0, accept the proposal.
- For all project with PVR < 0, reject the proposal.
- If PVR = 0, the projects is self-sustaining, i.e. it makes neither profit nor loss.
- If all projects have PVR > 0, accept the investment with the highest PVR.
- PVR is used for capital allocation because the project with the highest, PVR is most profitable.

Payback Period

Payback period refers to the time or point in the business life when the investor begins to make profit on his investment. It is also referred to as the time when the net cash flow turns positive, also refers to the number of years, months, hours, days etc. for the net cash flow to turn positive or the number of years for the investment to break-even as it is also known as "Break-even Period". It is calculated from the net cash flow using two methods, namely;

1) Graphical method
2) Trial and error method (interpolation)

Graphical method is a plot of the net cash flow Vs time.

Cash Flow Model

Cash flow model is referred to as the cycle of cash receipts (inward flow of cash) and cash disbursement (outflow of cash) that moves in or out of the treasury in a capital investment project (Iledare and Fubara, 2017). Cash flow models are commonly used to evaluate the profitability of a capital investment project. The cash flow is usually preferred for economic analysis because it employs discounted cash flow concepts that take into consideration the timing of cash flows and the economic opportunity cost on capital. Net cash flow (NCF) is simply receipts less disbursement during a business year usually annually and projected over the economic project life, which about 25 years for an oil and gas business.

Mathematically;

Net Cash Flow (NCF) = Net Annual Cash Receipts – Net Annual Capital Disbursements.

The discounted cash flow model equations under a concessionary (royalty and tax) and a contractual (production sharing contract) fiscal arrangement are represented as:

$$NCF = Gross\ Revenue - Royalty - Capital\ expenditure - Operating\ expenditure - Bonus - Taxes - other\ costs \quad (5)$$

Fiscal Incentive for petroleum industry fiscal bill 2017			
Fiscal incentives	Terrain	Classification	Rate (%)
Royalty by volume	Onshore	0-10	5
		>10≤20	15
		>20	20
Royalty by value	Threshold product price ($)		Rate (per $1 increase in price)
	50		0.2
Royalty by value (proposed for scenerio analysis)	35		20%
National hydrocarbon tax			30% of taxable income
Corporate income tax			30% of taxable income
Educaton tax			2% of taxable income
Nddc levy			3% of taxable income
Cost recovery limit			80% of net revenue after royalty
Profit oil share			To be negotiated

Above is an example of a cash flow model (A Nigerian fiscal system) (Nwosi-Anele et al, 2018)

Figure 2: Nigerian cash flow model

Nigeria

Overview of the Oil and Gas Industry

Nigeria with a population of about 172 million people is the most populous Black Country within OPEC and the world (OPEC 2015). Nigeria, which covers an area of about 924,000 km, is located in the Gulf of Guinea. The genesis of the oil and gas industry in Nigeria can be traced back to the early 1900s when a German company operating in the country began exploratory activities in the Araromi area, West of Nigeria. After several efforts and loads of investments, Nigeria got her first commercial discovery in 1956 at Oloibiri in Bayelsa state in the Niger Delta region (NNPC 2015). Nigeria has since grown to be the largest producer of oil in Africa having the largest reserves of natural gas in the continent. She ranks among the top five exporters of Liquefied natural gas (LNG) in the world (EIA 2015). The Nigerian oil and gas industry constitute upstream, downstream and midstream. The oil and gas industry in Nigeria which accounts for 75% of Government revenue and about 95% of total export revenue is the backbone of the economy. This has pushed other industries such as agriculture, which was the main revenue stream of the economy in the early fifties and sixties to the background (NNPC 2015).

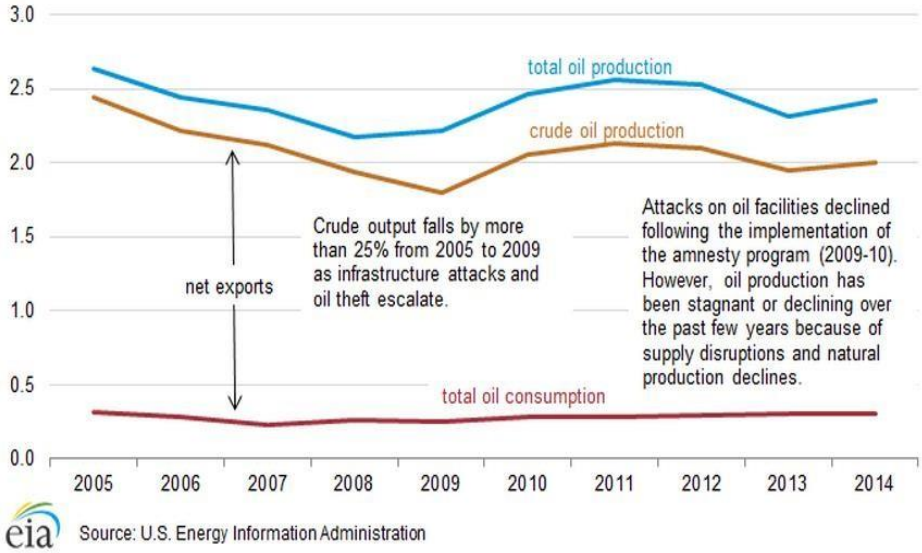

Figure 3: Hydrocarbon and consumption history in Nigeria (EIA, 2015)

Nigeria currently has an estimated 37 billion barrels of crude oil reserves which is the second largest after Libya (EIA,2015), majority which are found along the country's Niger river delta and offshore in the Bight of Benin, the Gulf of Guinea, and the Bight of Bonny. As at 2014, Nigeria produced about 2.4 million bbl/d of petroleum and other liquids of which 2 million bbl/d was crude oil while the remainder was composed of natural gas plant, liquid, condensate and refinery processing gains. Despite this large production rate and the presence of four oil refineries with a combined refining capacity of 445,000bbl/d, Nigeria still imports petroleum products to meet its local consumption demand which is about 305,000bbl/d. this is due to the fact that these refineries are working at only 22% of their working capacity (EIA 2015).

Figure 4: Map showing Nigeria and neighboring countries

Management Structure

It is of general experience that major IOCs tend to use sophisticated accounting packages to water down their rents such as taxable incomes though falsifying invoice, transfer pricing and other manipulative practice. These IOCs sometimes have their head offices in countries or regions such as Bermuda, British Virgin Island, etc where tax is not really taken in the strict sense of it (Nair 2012). This effectively creates an avenue whereby revenues generated from the Host countries are siphoned to other countries. To mitigate this trend, Host Governments have created several agencies and authorities to supervise the activities of this industry.

Nigeria is no different as it has a number of agencies tasked with various duties pertain the activities in the oil and gas industry. The 1991 Nigerian Constitution as amended gives the Government at the federal level sole ownership and control of all-Natural resources in the country, territorial waters and exclusive economic zone (Akanji 2010). Federal Government's powers include export duties, incorporation and regulation of corporate bodies, mines and minerals (including oil fields, oil mining, geological surveys and natural gas) and taxation of incomes, profits and capital gains. According to Babajide (2014), the Nigerian Government, the IOCs and indigenous companies found primarily in the service and downstream sector are the major players in the oil and gas industry. The Nigerian government focuses primarily on the upstream sector, the Nigerian National Petroleum Corporation (NNPC) with other supporting ministries such as the department of petroleum resources (DPR) (regulatory agency for oil and gas activities), the Ministry of Energy (MOE), the Federal Ministry of Environment (FME), the Federal Inland Revenue Service (FIRS). The Niger Delta Development Commission (NDDC) coordinates its activities. Other regulatory agencies include Ministry of Petroleum resources, Nigerian Investment Promotion Commission (NIPC), National Maritime Administration and Safety Agency (NIMASA), Nigeria Custom service Board (NCSB), Nigerian Content Development & Monitoring Board (NCDMB), etc. (KPMG 2014).

Nigeria Fiscal System

According to Ernest and Young (2014) and Babajide et al., (2014), two main types of fiscal systems exist in Nigeria: Concessionary and Contractual systems. Under the concessionary system, the government operates joint ventures with either an IOC or a sole risk operator while the contractual system involves PSC and the Risk sharing contracts (RSC). For a company to carry out petroleum operations in Nigeria, it is taxed under the

Petroleum Profit Tax Act (PPTA) 2004 (as amended). The term Petroleum operations as defined by the PPTA is the winning or obtaining, and the transportation, of chargeable oil in Nigeria by or on behalf of a company for its own account (Ernest and Young 2014). Arrangements involving the RSC are deemed not to be involved in petroleum activities rather on performance schemes and as such are taxed under the company's income Tax at a lower rate than the PPTA. For all activities involving oil and gas in the country, the NNPC operates on behalf of the Federal Government. Most of Nigeria's oil and gas projects are carried out through joint ventures between IOCs and the NNPC who hold majority shares (EIA, 2015).

Petroleum Industry Fiscal Bill (PIFB)

The proposed Petroleum Industry Fiscal Bill 2018 is a production sharing contract. It is a modification of the other versions of PSC with an aim to reduce tax burden on the contractor (Adaobi et al, 2018). Unlike other fiscal regimes, PIFB 2018 separates tax deduction of oil and gas operation, allowing oil exploratory and production activities to stand alone for tax deduction. Tax is deducted from the contractor's profit oil share. The profit oil is the net revenue after cost recovery that is split between host government and contractor at negotiable percentages but in previous fiscal regimes, it is shared between host government and contractor in the ratio of 70% to 30%. However, the contractor is to be taxed on his share of the profit oil. A tax shall be levied upon the profits of any company engaged in upstream petroleum operations known as Petroleum Income Tax, which shall be charged, and assessed upon its profits and payable during each Accounting Period in accordance with the provisions of this Act (PIFB, 2018). The assessable tax for any accounting period of a company shall be a percentage of the chargeable profit for that period aggregated separately as follows:

(a) 65% for onshore areas crude oil profit
(b) 50% for shallow water areas crude oil profit

Table 1: Royalty deduction

ONSHORE		**SHALLOW WATERS**		**OFFSHORE**	
TRANCHES (KBPD)	ROYALTY RATE %	TRANCHES (KBPD)	ROYALTY RATE	TRANCHES (KBPD)	ROYALTY RATE
2.5	2.5	First 10	5	First 50	5
7.5	7.5	Next 10	10	Next 50	7.5
10	15	Next 10	15	Above 100	10
Above 20	20	Above 30	20		

Overview of Petroleum Prospects

Nigeria currently has 606 oil fields in the Niger Delta of which 355 are on-shore while 251 are offshore. About 193 fields are in operation currently while 23 are shut down, 28 exploratory wells have been drilled outside of the Niger Delta having varying level of prospect. These include one discovery well in Edo state and Benue State each, 2 discovery wells in Anambra and about twenty- four in the chad basin (NNPC 2015).

Angola

Overview of the Oil and Gas Industry

The Republic of Angola having a population of about 19 million, a GDP of $6,282 per capita and landmass of 1.25 million square km is located on the Atlantic coast in the Southern part of Africa and is the second largest oil producer in Africa (EIA 2015, OPEC 2015). Exploratory activities in Angola began in 1910, however drilling commenced commercial in 1956 (Rosenblum, 2011). Since this discovery, offshore fields have been the dominant force accounting for majority of production capacity (EIA 2015). Angola became a full member of OPEC in 2007, assuming the presidency of the organization 2009 and has a current production quota set at 1.65 million bbl/d (Brown et al., 2015). Oil exploration is mainly done offshore at depths of over 1,200 meters (Sonagol 2015). As shown by EIA (2015) and Brown et al., (2015), the oil sector contributes about 45% of the nation's GDP, constituting 95% of total exports while driving the economy at an impressive 5.6% growth rate.

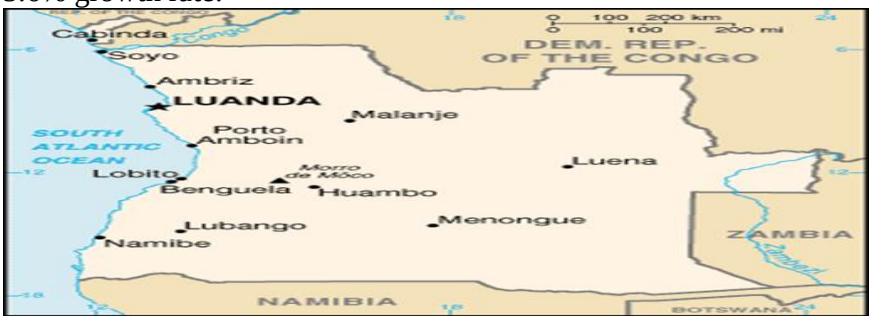

Figure 5: Map of Angola and neighboring countries

Angola currently has a proven oil reserve of 9 billion barrels with current production capacity at 1.7 million bbls/d and refinery capacity of about 45,000 bbls/d (OPEC 2015). It also produces light sweet crude making its crude highly marketable in the global market. Current markets for Angolan crude are include China, USA, The EU and India although a decline export to the USA has reduced drastically due to the oil glut in the USA (EIA 2015).

Management Structure

Sociedade Nacional de Combustíveis de Angola (Sonangol) which was created in 1976 is Angola's national oil company, the administrator of mineral rights and the centre of the petroleum industry saddled with administrative duties in relations with oil industry operators both international and local (McHarg 2010). Sonangol with 17 subsidiaries has major shares in almost all of the 44 oil and gas production and exploration blocks in the country (EIA 2015). By law Sonangol operates joint venture partnerships and production sharing contracts with International Oil Companies (IOCs), with Chevron, Total, ExxonMobil, BP, Statoil and Eni among others forming major partners (EIA 2015). The 2 most important pieces of legislation governing oil and gas operations in Angola are The 2004 Petroleum Activities Law and the 2004 Petroleum Taxation Law establishing the state's control over mineral rights and Sonangol's role in the management of the operations of the oil and gas sector (Taverne 2008). Angola also has a local content policy

that regulates IOCs to meet a 70% Angolanization threshold, fund technical training programs and use local banks for all their transactions (EIA 2015).

Angola Fiscal System

There are currently three main types of fiscal systems in Angola operating under different tax regimes (Ernest and Young 2014); Risk service contracts (RSC), Partnership that is only applicable to certain agreements set up in the 1960s and 1970s and Production sharing agreements (PSA) which is the most common form of agreement. According to EIA (2015), IOCs currently operating in the country or are intending to operate must operate under Joint- ventures and PSA with Sonagol. Examples of some joint venture companies with Sonagol include Sonils (Provision of Logistical support for oil exploration and service companies), Angloflex (building umbilical and associated products for the oil and gas industry) and Sonatide (provision of marine services to oil operations) (Sonangol 2015).

Overview of Petroleum Prospects

Currently, Angola has about 44 oil blocks both onshore and offshore of which 33 are in exploratory mode and 11 in production mode (EIA, 2015). According to Sonagol (2015), commercial discoveries in the deep waters consisting of block 17, 14, 15 and 16 marked the breakthrough in the Angolan oil production history.

Algeria

Overview of the Oil and Gas Industry

With a land area of around 2,382,000 Sq km, the Republic of Algeria with a population of about 38 million people is the largest member country of OPEC territorially (OPEC 2015). It shares its borders with Morocco, Mali, Libya, the Western Sahara, Tunisia and the Mediterranean Sea to the North (OPEC 2015). The first commercial discovery of oil was at Edjelleh in 1956 with the HassiMessaoud oil field following suit in the same year. Production of oil began in 1958 (OPEC 2015).

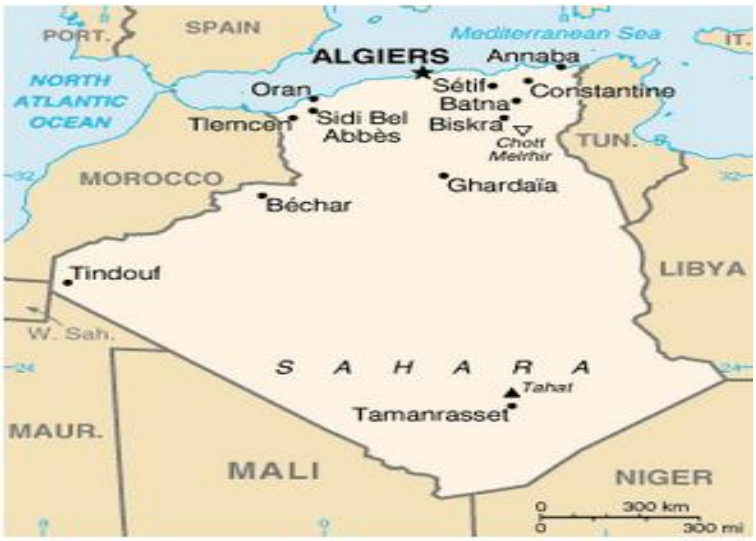

Figure 6: Map of Algeria and neigbouring countries

As of 2014, Algeria's crude oil reserves stood at 12.2 billion barrels with daily production and consumption set at 571,400 bbl/d and 380,000bbl/d respectively (CIA 2015), thus making it the third largest proved crude oil reserve in Africa. All of which are offshore due to the increasing rate of Militants groups in North Africa and the Sahel (EIA 2015). It also produces high quality crude with low sulphur content. Natural gas production and consumption was estimated to be at 78.6 billion cu m and 32.3 billion cu m as at 2013 making it the 10th largest natural gas holder in the world (EIA 2015). Algeria has five oil refineries with a refinery capacity of about 652,500 bbl/d with Skikda located along the Algerian coastline been the largest but locally and the whole of Africa with a refinery capacity of 352,700 bbl/d of crude oil and condensate (EIA 2015).

Management Structure

Algeria's oil company, Entreprise Nationale Sonatrach which is the largest oil and gas company in Algeria and Africa owns approximately 80% of total crude oil production in Algeria while the remaining 20% is owned by IOCs working in Algeria (EIA, 2015). The hydrocarbon act of 1986 and 2005 governs the Oil and gas industry (Ernest and Young 2014). The 2005 law, which was amended in 2006 to be more favourable to IOCs, grants Sonatrach a minimum equity stake of 51% in all hydrocarbon projects. (EIA, 2015) reports that due to difficulties in attracting foreign investors the hydrocarbon law was revised in an attempt to attract foreign investors to upcoming projects. The amendment does not change Sonatrach's position as it continues to be the majority stakeholder in all upstream oil natural gas projects (EIA 2015).

Algeria Fiscal System

As mentioned earlier, all issues pertaining to matters relating to the upstream oil and gas industry in Algeria is governed by the Hydrocarbon act of 1986 and 2005 (Ernest and Young 2014.) This been said, different fiscal systems apply in Algeria depending on the period, which the contract was signed (Agalliu 2011, Ernest and Young 2014). The 1986 which comprises of royalties, income tax and tax on extraordinary profit (TEP) or windfall tax, governs the appraisal to the transportation phase, while the 2005 law which comprises royalties, PIT, Additional profits tax (APT) and surface tax entails refining to the distribution of petroleum products with installation sites that enable these processes. The law grants Sonatrach a minimum equity stake of 51% in all hydrocarbon projects as they maintain a key role in all contractual agreements.

Overview of Petroleum Prospects

(EIA 2015) shows that according to recent analysis, no major crude oil project is been scheduled to take off soon in Algeria. Currently, there is a decline in the production rates in older fields in Algeria, as such; smaller projects such as the BirSeba and Timimoun projects are scheduled to come on stream soon to bolster the production rate in existing fields such as Gassi Touil-Rhoude Nouss and Hassi Messaoud. All these have lead the EIA to project that crude oil output in Algeria will gradually decline in the short and medium term. A report by (Coats 2014) shows that he Algerian Government plans to invest about $100 billion into reviving the country's energy sector up until 2020. This would be done mainly by increasing exploration efforts, increasing local reserves and boosting the county's new shale plans.

Libya

Overview of the Oil and Gas Industry

Situated in the North of Africa and sharing a border to the West with fellow OPEC member country Algeria, Libya is 16th largest country in the world in terms of land mass comprising around 1760km^2 (OPEC 2011). In 1959 the first successful drilling was reported when Zletin oil field, one of the largest in the country was discovered by Esso Libya (Libya oil almanac, 2011). With the discovery of significant oil reserves in 1959, Libya moved abruptly from being independent on international aid and the rent from US and British air bases to being an oil-rich monarchy (Encyclopedia Britannicca, 2011). Libya joined OPEC in 1962 (Frank and Francis, 1980).

Apart from petroleum Libya's natural resources are natural gas and gypsum, its economy depends primarily on the oil sector which report about 82 percent of export earnings. Moreover, the oil and gas sector accounts for 70% of its GDP. Substantial revenues from the energy sector coupled with a small population gave Libya one of the highest per capita GDP in Africa (OPEC 2012). Libya produces an average of 1.7million barrels daily (OPEC, 2012).

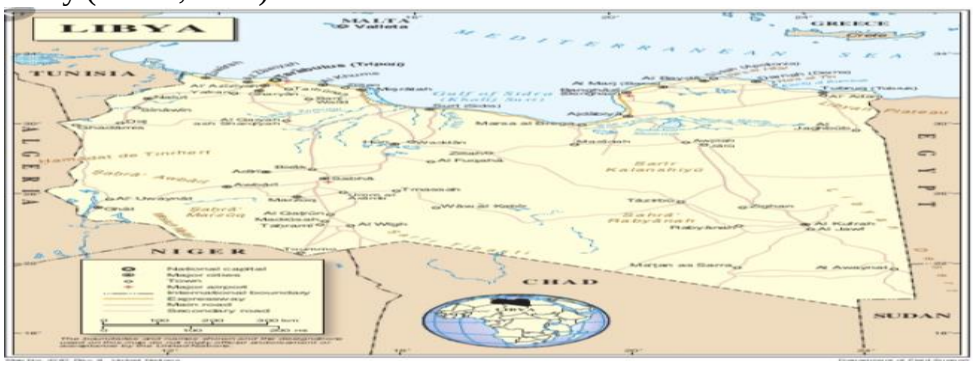

Figure 7: Map of Libya and neighbouring countries

Management Structure

Libya's oil industry is run by the state owned National Oil Corporation (NOC) which is the body responsible for implementing the Exploration and Production Sharing Agreement (EPSA) with IOC. The NOC is also responsible for field development and improvements as well as downstream activities. IOC participation in Libya's oil concession was initially as high as 49%. However, the changes to the production sharing agreement under the EPSA IV Licensing Round in 2005 limited IOC production shares. The Libyan government has since required that IOC's already operating in the country rewrite existing contracts to comply with the new framework (Libya documentary, 2011).

Libya Fiscal System

The Petroleum Law No.25 was enacted in 1955 which provided a legal basis for the oil industry and precipitated a period of rapid growth in exploration activity. After the enactment of the petroleum law, the Libyan government conducted numerous renegotiations

of concessions previously offered under the original petroleum law and managed to impose tough fiscal terms on IOC using favorable market conditions to its advantage (Clyde and Co, 2013). In 1972 participation agreement replaced the concessions transferring 51% of all concession to the Libyan National Oil Company. In 1973, the Libyan government introduced (EPSA). Ten (EPSA I) contracts were signed in 1974-1979 period. The government introduced EPSA II in 1979 to reduce concerns over reserve replacement ration. Faced with the collapse of oil price in mid-1980, low internal production, the effect of the economic sanctions and financial difficulties linked to considerable expenditures on the Great man-made river project and Missrada steel works, the government introduce the EPSA III framework in 1988 with improved terms in order to attract investors. The EPSA III framework featured significantly higher production shares for IOC compared to the later EPSA IV framework introduced which entailed higher Libyan government shares. EPSA IV is the most recent contractual model used by NOC to govern contracts with IOC operating in Libya (Libya Documentary, 2011).

Overview of Petroleum Prospect

Libya is back after production oscillated from 1.6 million barrels per day before their revolution to as low 200,000 barrels per day. As major IOCs return to Libya, recovery should gather pace. The return of Schlumberger to work with the Sirte company for oil and gas production, a unit of the NOC; of Winterstall and Gazprom, which have restarted production from the Asarah field in the Sirte basin of Eni which has bought new wells online at its Bahr Essalam development and the NOC's agreement with Shell to provide oil for the rest of 2018 have helped stimulate other participants (Francis Ghiles, 2018). Total has announced it will buy marathon's state in the Waha and British Petroleum and Eni announced plans to begin exploration in the Ghadames Basin in western Libya.

Software Module

Microsoft Excel: has the basic features of all spreadsheets (Harvey et al 2006), using a grid of cells arranged in numbered rows and letter-named columns to organize data manipulations like arithmetic operations. It has a battery of supplied functions to answer statistical, engineering and financial needs. In addition, it can display data as line graphs, histograms and charts, and with a very limited three-dimensional graphical display. It allows sectioning of data to view its dependencies on various factors for different perspectives (using pivot tables and the scenario manager). It also has a variety of interactive features allowing user interface that can completely hide the spreadsheet from the user, so the spreadsheet presents itself as a so-called application, or decision support system (DSS) via a custom-designed user interface, for example, a stock analyzer (Şerefet al, 2008), or in general, as a design tool that asks the user questions and provides answers and reports (Wells et al, 1997). Ms excel will be used to develop a cash flow model. The cash flow model will be used for the calculation of the government's take and contractor's take economic tool analysis.

Methodology

This section shows the method by which the data were collected, collated and analyzed in achieving the set aim and objectives of the project. With the use of both qualitative and quantitative methods, while relying on secondary data source, the aim of this work will be achieved.

Compendium of the Hypothetical Field

The XX field is situated on an anticline and positioned between two major growth faults. The field is in a shallow offshore environment in the YY area. The XY reservoir contains about 90% of the field's reserves. Primary production from these reservoirs is by gravity drainage and gas-cap expansion. The effect of water influx on the total recovery mechanism has been minimal. Reservoir engineering studies indicate primary depletion will permit the recovery of 45% of the original oil in place (OOIP). The XX field was discovered in 1992. The field is a shallow offshore with water depth of 40m. The XY reservoir is compositional and heterogeneous nature with neighboring reservoirs.

Data Collection

Data collection according to the business dictionary is "the collection of data, from survey or from independent or networked locations via capture, data entry or data logging". As stated earlier, secondary sources bordering around financial analysis were used for data collecting data about this research. The researcher collated data sets from the various case study countries which were readily available from verified sources such as reliable websites, journals, articles, Government annual reports, Multinational oil companies reports and conference papers. All these were done making justifiable and feasible assumptions where needed. Major sources of data for this research are listed below:

- Nigerian Ministry of Petroleum
- EIA
- OPEC
- World bank
- Nigerian National Petroleum Corporation
- Angola Ministry of Petroleum
- Algeria's National Oil company, SONATRACH
- Angola's National Oil company, SONAGOL
- Libya's National Oil Corporation
- Online Publications
- Journals and Articles

Research Design

In order to evaluate the comparative performance of the fiscal regimes of four countries namely; Nigeria, Angola, Algeria and Libya respectively. A combined technique of both Qualitative and Quantitative analytical methods was employed to justify the research objectives. Quantitative Analysis involves determination of Fiscal terms for case study countries and determination of input variables assumptions. Qualitative Analysis involves management structures for case study countries.

Data Analysis

For the purpose of this research work, various assumptions were made concerning field data. This analysis is carried out based on a hypothetical field. For comparison purpose, similar production profile has been used for the purpose of this analysis with only the fiscal system for each case study country differing in the analysis. In addition, price of oil as at 9 August 2019, which is, $57 is used as the base case for the analysis. Table 2 below shows the basic input variables and assumptions used for the purpose of the data analysis for this research.

Table 2: Field Data

Parameters	Values
Field type	Shallow Offshore
Initial Oil price	$57.00
First Production	130,000bbls
Peak Production	250,000bbls
Build-up period	5 years
Plateau Period	6 years
Decline Period	14 years
Interest rate	10%
CAPEX	$1,958,910,000.00
Cost of asset	$511,451,000.00
Salvage value	$11,806,000.00

From the production profiles, the estimated field life of the hypothetical field is approximately 25 years. As such, it is projected to commence production by reaching its peak production in 2020.

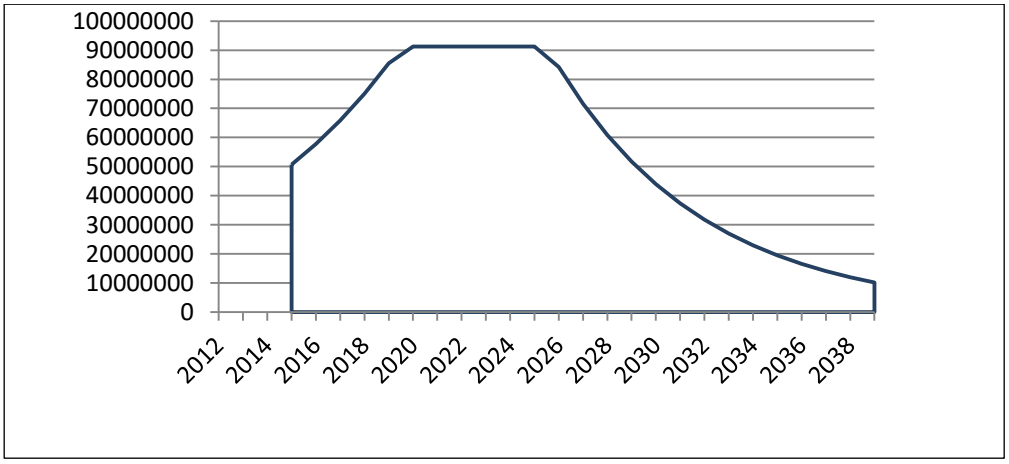

Figure 8: Project production profile

The input variables considered in this work for the respective countries will be outlined below in table 3.

Table 3: Fiscal System Features

NIGERIA	
Fiscal regime	Petroleum Industry Fiscal Bill (PIFB)
Royalty	20% (due on the gross income)
Petroleum Income Tax (PIT)	50% for shallow water areas crude oil profit.
Profit share	Gtake 60% and IOC 40%
Signature bonus	$250,000,000.00
ALGERIA	
Fiscal regime	Royalty/tax
Royalty	20% (due on the gross income)
Income Tax	38% levied on the profit made by the foreign partner or a sonatrach (NOC in Algeria
TEP (Tax on Extraordinary Profit)	15% - It applies to the foreign partner when the average oil price exceeds $30/bbl ranging from 5% to 50%.
ANGOLA	
Fiscal regime	Profit Sharing Contract
Royalty	None
Training tax	• $100,000 yearly, companies with a prospecting license • $300 yearly, companies with an exploration license. • $0.15/bbl for production companies and also engaging in refining and processing
Petroleum Income Tax (PIT)	50%
LIBYA	
Fiscal Regime	Exploration Production Sharing Agreement
Signature Bonus	$250,000,000.00
Royalty	16.75% (due on the gross income)
Corporate Income Tax	20%
Jihah tax	4%
Surtax	41%

Research Constraints

This research work was subjected to a wide variety of online data sources and information but time and resources were a constraint for the whole of the research. However, the use of primary data would have added more authenticity to this research but a more time friendly method of data collection was used. Efforts were however made to verify and justify the data sources and data collected. Lack of transparency in the data released by the various case study government and IOCs and other sources used led the researcher to make well documented assumptions where needed. These assumptions made may not totally represent current situations in these countries and as a result presented its limitations coupled with the fact that IOCs and each country's Government may not release full contractual agreement to the public but just model contracts.

Discussion of Results

This section presents the analysis and results of the various case study fiscal regimes in terms of ways by which profits are shared from revenue generated between the IOC and Government in each fiscal system. Also, analysis would be carried out on the Strength, Weakness, Opportunity and Threats of each country which is in accordance

with the project objectives set out in chapter one, which shall constitute a basis by which a very comprehensive comparison can be made for intending foreign or existing investors.

Comparison of Fiscal Systems in (Nigeria, Angola, Algeria and Libya)
Economic Tools Analysis

Table four shows the breakdown of the various economic parameters, which have been analysed using the various data and assumptions shown in table 2.

I. Net Cash Flow (NCF)

The importance of the NCF of a proposed project cannot be over emphasised as it forms the basis by which stakeholders (IOC/NOC) can compare the estimated earnings that can be generated from the project (Govindaraian 1980). For this purpose, the Before Tax Net Cash Flow (BTNCF) and After Tax Net Cash Flow (ATNCF) projections of the various case study countries have been analysed. From the table below, the BTNCF for Nigeria, Angola, Algeria and Libya are $23.1billion, $29.6billion, $60.7billion and $17.3billion respectively.

Table 4: Summary of Economic Indicators

	Nigeria	**Angola**	**Algeria**	**Libya**
	PIFB	PSC	ROYALTY/TAX	EPSA
BTNCF ($MMM)	23.1	29.6	60.7	17.3
ATNCF ($MMM)	11.5	13.0	26.8	6.0
NPV ($MMM)	2.3	3.45	8.3	3.5
PI	2.168	2.763	5.248	2.792
PVR	1.168	1.763	4.248	1.792
IRR (%)	37.92	42.32	48.22	48.19
PAYBACK PERIOD (Years)	7.9	6.8	5.1	5
GOVT TAKE (%)	91.56	72.78	85.68	90.66
CTAKE (%)	8.44	27.22	14.32	9.34

Source: Author's compilation

However, for the purpose of this comparison and to determine how profitable this project is to the IOC, the ATNCF and DNCF have been focused on by the author. While undiscounted ATNCF does not show the relationship between cash annual cash flow and the time value of money (Drury 2008), discounted ATNCF or NPV considers this and as such is used as the basis for comparison. ATNCF for the case study countries is shown in table 4 above with Libya having the lowest ATNCF basically due to its compulsory 65% tax together with the high Government 70% profit oil share. Algeria has the highest ATNCF as it operates a Royalty/Tax fiscal regime with only the royalty of 20% and income tax of 38% at play in the country and factored into this analysis. The implication of this is that, at the current price of $57, an average field in Algeria is still viable. This applies to the other case countries too but with smaller values.

II. Net Present Value (NPV) and Payback Period (PP)

As shown in table 4 above, the various case study countries all show positive NPVs. This is the sum of the individual yearly NCF after been discounted using a discount rate of 10% as compared to industry standards. This implies that these various countries if invested in, have the potential of making profit for the stakeholders involved. The NPV of any project takes into consideration the time value of money, which is an important criterion for which investors make informed decisions while payback period is the number of years it takes for a project to start yielding profit. Nigeria's projected NPV of $2.3 billion is the lowest of the four countries with a payback period of about 7.9 years with Angola, Algeria and Libya having $3.45, $8.3 billion and $3.5 with payback period of 6.8 years, 5 years and 5.1 years respectively. All positive NPV shows the wealth generating potential of this project using the different fiscal regime parameters of the different case study countries.

The figure below shows the estimated projections of the annual DNCF for the various countries. As the NPV of any projects relates to stakeholders' wealth (IOC most particularly in this case), Algeria having the highest NPV is the most profitable when stakeholders plan to base investment decision on this economic tool with a short payback period of 5 years. As shown in the figure Nigeria is the least attractive of the four based on the analysis as its NPV of $2.3 billion together with the long time it takes for it to start making profit. It is the lowest of the four countries owing mainly to the huge PIT that runs at 50% taxable income as stipulated in its fiscal laws.

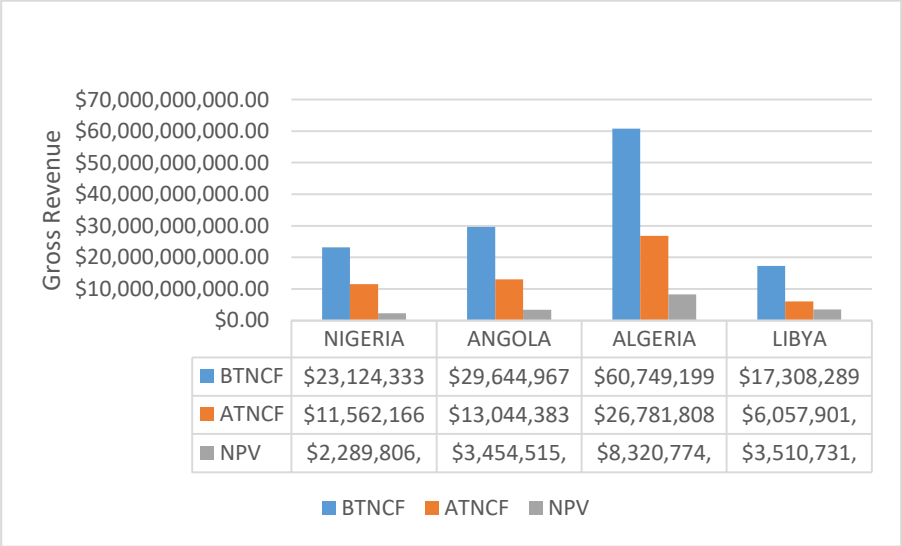

Figure 9: Comparison of BTNCF, ATNCF and NPV

III. Profitability Index (P/I)

For a project to be deemed profitable, P/I must be greater than 1.0 as a ratio lower than 1.0 indicates that the project would lose money. Nigeria has the lowest value of 2.168, while Angola, Algeria and Libya have 2.763, 5.248 and 2.792 respectively. This implies that for every one dollar invested in each of the case study countries, there is a profit of 1.168, 1.763, 4.248 and 1.792 to be gained with Algeria been the most attractive.

IV. Internal Rate of Return (IRR)

The IRR was calculated using the trial and error method. From table 4 it can be seen that the IRR for Nigeria, Angola, Algeria and Libya are 37.92%, 42.32%, 48.22% and 48.19% respectively. Generally speaking, projects with IRRs greater than the capital cost are viable. Any rate below these IRR rates for any of these countries would render the project non-viable. From the analysis done for the purpose of this research, Algeria and Libya show near similar IRR compared to that of Nigeria and as such make more economical sense.

V. Government CTake/ IOC Take (CTake)

From our analysis as shown in table 4 and for the purpose of our comparison, CTake for Angola 27.22% is the largest followed at a distant by Algeria, Libya and Nigeria respectively with CTake values of 14.32%, 9.34% and 8.44% respectively. This implies that of all four case study countries, those of Algeria, Libya and Nigeria have tougher fiscal terms which demand more economic rent for the HG making Angola the most attractive for IOCs based on the "TAKE' analysis. Figure 4.4 shows the graphical representation of these different scenarios.

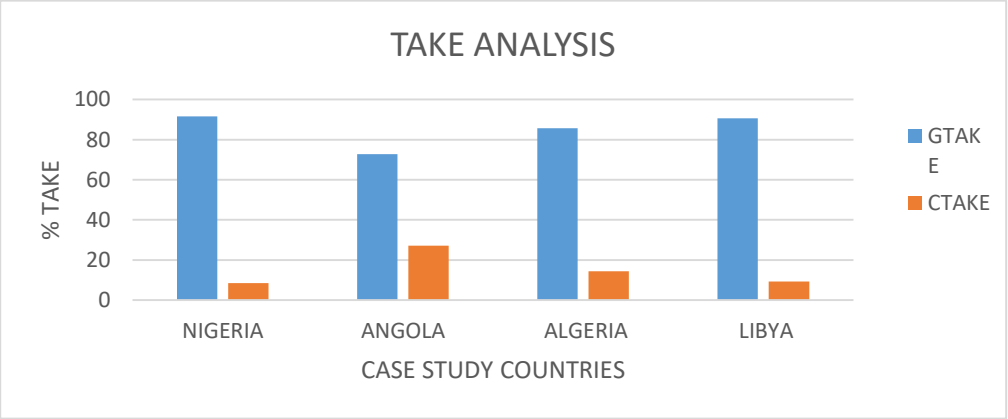

Figure 10: Take Analysis

In an attempt to back this quantitative comparative analysis of the four case study countries and according to the methodology set out for this research work in order to achieve objective two, a SWOT analysis table is shown below. These would help bring to perspective the various happenings in these countries.

Table 5: SWOT Analysis of Case Study Countries in Terms of Economic Viability

	Nigeria	**Angola**	**Algeria**	**Libya**
Strength	12th largest crude oil producer in the world, contributing about 3% of the global crude oil	Proven substantial oil reserves in tunes of about 13.5 billion barrels making it a key player in Africa's oil industry as both a major	substantial oil and gas reserves and production. This sector contributes over 50% to GDP and more than	The holder of Africa's largest proven oil reserves of 48.01 billion barrels. Also it is one of Europe's biggest North

	production thus making it the largest crude oil producer in Africa. The largest gas reserve in Africa and also contribute about 8% of the global liquefied natural gas supply. Nigerian crude oil is a high quality, light low sulphur grade referred to as sweet, which is highly prized by oil consuming nations because of its high gasoline content and relatively cheap processing costs. Large population of about 150 million people. Production of more than 2.8 million barrels per day with a total of over 159 oil fields and over 1481 wells in operation for exploration and production of its crude oil and natural gas resources.	producer and exporter. Daily crude oil production of about 1.65 million barrels per day with its crude oil exports accounting for up to 1.61 million barrels per day. The country's geological endowment is so favourable and thus the petroleum sector is seen as relatively stable, attracting more investment from foreign multinational oil companies. It accounts for over 40% of GDP, over 90% of export revenues and 80% of government revenues thus making it a major generator of funds to aid Government financial spending.	96% to government revenues in the country. It has an oil refining capacity of 650,860 bpd although the actual refined petroleum output is about 500,000 bpd	African oil suppliers. It supplies from North Africa to Europe destinations with advantage of being booth timely and cost effective. Strategically located on the Mediterranean near Europe. The oil industry contributes to 70% of the Gross Domestic Profit.
Weakness	Required infrastructure to make the natural gas useful are not available; hence, it is mostly flared away making it the second worst country in terms of gas flaring in the world. Pipeline vandalism, oil theft, militancy, frequent changes of the	The oil industry is dominated by the upstream sector while the downstream sector remains underdeveloped. With the administration and regulation of the oil and gas industry in Angola solely in the hands of the state-owned oil company, Sonangol, the company's transactions with the national budget are porous and allow	Restrictive government regulations cause decrease in foreign direct investment (FDI) and ownership of stocks by foreign companies in the petroleum sector. Decline in the nation's production of natural gas due to mismanagement of the sector as well problems of insecurity and high militancy	There is no clearly defined industrial strategy. Decline in the nation's production of oil due to problems of insecurity and high militancy

	management of the NNPC, the organisation in charge of the sector, ageing assets and also the volatility of oil prices are pitfalls presently befalling the Petroleum sector.	for funds to be easily siphoned off breeding corruption Insufficient infrastructure for the distribution of revenue generated from the sector		
Opportunity	Creation and establishment of more Liquefied Natural Gas (LNG) plants in order to harness the massive gas reserves the industry explores and produces. The deepwater sector of the industry still has a large avenue to expand and develop as it is less disturbed by local militants' attacks, seizures due to conflict and sabotage.	The reorganization of the sector with the framework of the new Petroleum Activities Law attracted significant investments from foreign companies has contributed to the significant increase in production.	substantial unexplored shale gas resources thus encouraging its aims to develop renewable energy sources. Recent boast in the Oil and gas sector through partnership with Foreign multinational companies to facilitate a shift from a state owned sector to a private led one.	Establishment of a pipeline between Egypt and Libya as Egypt is upgrading its refineries.
Threats	Activities of militants concentrated along the Niger-Delta region remains a potential threat to the sector. Corruption and misappropriation of petroleum funds realised from the industry. Oil theft in the Niger Delta.	Corruption and public-sector mismanagement. Falling oil prices globally also threaten the growth of the oil and gas industry in Angola	The rate of exploitation of this sector's resources outpaces the discovery of new reserves. As a result of the civil war (1992 – 2000) military influence, corruption and cronyism remain a very big threat to the growth of the sector bringing about laundering and pilfering of funds realised from the petroleum sector The issue of terrorism/ radical jihadist groups presents a terror threat to the petroleum sector.	Unfavourable business opportunities. Military control of the oil crescent has changed hands though fighting continues near the terminals and oil export could be hit. Unstable government caused a drop in the production of oil.

Conclusion

The main challenge for the harnessing of Crude oil resources is embedded in making the right decisions and the careful and strategic implementation of these decisions to meet IOC or HG objectives while at the same time meeting stakeholders' growing demands. As such, this study is important as it relates directly to IOCs in their quest to expand operations while looking for markets that allow for the successful repatriation of stakeholders' wealth. In this study, the competitiveness of PFS around the world is discussed with particular reference to Africa as it pertains to foreign investment using Nigeria, Angola, Algeria and Libya as case studies with the aim of comparing the attractiveness of crude oil exploration and production in this region. For comparison, similar field size, production rate, CAPEX, and oil price have been used for all case study countries as was adopted by Echendu (2012) and Zahidi (2010).

It is relatively difficult to decide on the exact ranking of fiscal systems in terms of attractiveness as IOCs rank priorities of these criteria according to their suitability to profit profile. However, in order to analyse the profits to be gained from risk taken by these IOCs without taking into considerations these priorities, fiscal structure and take analysis have been carried out in this study. The following conclusions were made:

- A case flow model was developed for the purpose of this study for comparative attractiveness of Petroleum Fiscal Structure from 4 countries to be achieved.
- All case study countries had positive and impressive NPVs at the case oil price, with Nigeria having the lowest NPV
- Results obtained shows that the IRR for each country was above the threshold (Discount factor) making investment in these countries viable.
- On average, the payback period in the case study country is about 6 years from development with Algeria and Libya being the shortest, which is good indicator for investors.
- Angola has the highest CTake while Algeria has the highest NPV of the four countries.

Recommendation

After careful analysis of results, the following recommendations were made.

- Input variables should be refined to show real life situations instead of the use of hypothetical data in some cases.
- This research work should be broadened to PFS in other African countries.
- As stated earlier in this study, time was a limitation and such it is recommended that a longer timeframe should be given for this future studies.

References

Adamu, M. A., Ajienka, J. A., and Ikiensikimama, S. S. (2013). Economics Analysis on the Development of Nigerian Offshore Marginal Fields Using Probabilistic Approach. *Advances in Petroleum Exploration and Development*, *6*(1), 11-21.

AFDB (2015). *Industries and Services - African Development Bank* [online] available from <http://www.afdb.org/en/topics-and sectors/ sectors/private sector/areas-of-focus/industries-and-services/> [27 May 2015].

Amoako, J. and Owusu, J. (2010). *An Evaluation of Ghana's Petroleum Fiscal Regime* [online]. Accra: The Ghana Policy Journal. available from <http://ieagh.org/wp-content/uploads/2014/07/gpj-v4-art1.pdf> [27 June 2015].

Babajide, N. A.a, Ogunlade, C. A., Aremu, D. O., Oladimeji, S. T., and Akinyele, O. A. (2014). Comparative Analysis of Upstream Petroleum Fiscal Systems of Three (3) Petroleum Exporting Countries: Indonesia, Nigeria and Malaysia**;** International Journal of Sciences: Basic and Applied Research (IJSBAR) (2014) Volume 15, No 2, 99-115.

Bastida, E., Wälde. T. & Warden-Fernández, J. (2005). International and Comparative Mineral Law and Policy – Trends and Prospects. The Hague, Netherlands: Kluwer Law International.

Blake, A. and Roberts, M. (2006). 'Comparing Petroleum Fiscal Regimes Under Oil Price Uncertainty'. *Resources Policy* 31 (2), 95-105.

Brown, M., Lawrence, J., Jankowski, P., Brinkley, J., Hughes, M., Strauss, K., Mendelson, S., McIntyre, H., Rapoza, K., News, K., News, K., Solomon, B., David, F., Rooney, J., Weissman, J. and Lariviere, D. (2015). *Angola - Forbes* [online] available from <http://www.forbes.com/places/angola/> [24 June 2015].

CIA, (2015). *The World Factbook* [online] available from <https://www.cia.gov/library/publications/the-world-factbook/geos/ag.html> [27 June 2015].

Coats, C. (2014). *Algeria Spending $100 Billion To Turn Around Falling Oil And Gas Production* [online] available from <http://www.forbes.com/sites/christophercoats/2014/07/24/algeria-spending-big-to-boost-production/> [29 June 2015].

Cunningham, M.T. (1986). *"Strategies for International Industrial Marketing"*. In D.W. Turnbull and J.P. Valla (eds.) Croom Helm pg. 9.

Dharmadji T. and Parlindungan T. (2002). *Fiscal regimes competitiveness: Comparison of Oil and Gas Producing Countries in the Asia Pacific regions: Australia, China, India, Indonesia and Malaysia*. Paper SPE 77912 presented at the SPE Asia Pacific Oil and Gas Conference and Exhibition held in Melbourne, Australia, 8 – 10 October.

Echendu J. C. (2012). *Comparative Performance Analysis of Petroleum Sharing Contracts in Angola, Equatorial Guinea, Gabon and Nigeria*. Paper SPE 162990 presented at the 2012 SPE Nigerian Annual International Conference and Exhibition held in Abuja, Nigeria 6-8 August 2012.

EIA, (2015). *Nigeria - International - Analysis - U.S. Energy Information Administration (EIA)* [online] available from <http://www.eia.gov/beta/international/analysis.cfm?iso=NGA> [25 June 2015].

Ernest & Young (2015). 'AFRICA: Investment Attractiveness Survey'. *Africa Research Bulletin: Economic, Financial and Technical Series* 52 (2), 20755A-20755C.

Ernest & Young (2019). *Oil Tax Guide*, 1-20, 382-389.

Govindarajan, V. (1980). *'The Objectives of Financial Statements: An Empirical Study of the Use Of Cash Flow And Earning By Security Analysts'.* Accounting Organizations and Society 5 (4), 383-392.

Johnston, D. (1994). *International Petroleum Fiscal Systems And Production Sharing Contracts*. Tulsa, Okla.: PennWell Books.

Johnston, D. (2003). International Exploration Economics, Risk and Contract Analysis. Tulsa, Okla: PennWellKaiser, M. (2007) *'Fiscal System Analysis—Concessionary Systems'*. Energy 32 (11), 2135-2147.

Kaiser, M.J. and A.G. Pulsipher (2004). *Fiscal system analysis: Concessionary and contractual systems used in offshore petroleum arrangements*. U.S. Department of the Interior, Minerals Management Service, Gulf of Mexico OCS Region, New Orleans, La. OCS Study MMS 2004-016, 78.

Kellas, G. and O'Rourke, C. (2006). *Angola'S E&P Fiscal Regime in a Global Context*.

Libya Government (2006). *Libya Model Agreement* 28 -34.

Libya Government (2006) *Libya Oil Almanac* 9-47

Lynley, M. (2011). United Nations Report: On renewable resources can outpace global energy demand by 2020. Retrieved from http://www.venturebeat.com on 26 June 2015.

Mazeel, M. (2010). *Petroleum Fiscal Systems and Contracts*. DiplomicaVerlag.

Mian, M. A. (2010). *Designing Efficient Fiscal System.* Society for Petroleum Engineers Committee Paper. USA, Dallas, Texas.

NNPC, (2015). *Industry History* [online] available from <http://www.nnpcgroup.com/NNPCBusiness/BusinessInformation/OilGasinNigeria/IndustryHistory.aspx> [25 June 2015].

Nwosi Anele A.S, Adeogun Oyebimpe and Iledare Omowumi (2018), *Analysis of Government and Contractor take statistics in the proposed petroleum industry.* SPE conference paper presented at NAICE Lagos, Nigeria. Paper No SPE-193470-MS.

Offshore-technology, (2015). *East Hub Development Project - Offshore Technology* [online] available from <http://www.offshore-technology.com/projects/east-hub-development-project/> [13 July 2015].

Onaiwu E. (n.d) How do fluctuating oil prices affect Government take under Nigeria's PSCs University of Dundee [Online] available from < http://www.dundee.ac.uk/cepmlp/gateway/?news=30868> [09 July 2015].

Opec, (2015). *OPEC: Nigeria* [online] available from <http://www.opec.org/opec_web/en/about_us/167.htm> [25 June 2015].

Ovadia, J. (2014). 'Local Content and Natural Resource Governance: The Cases Of Angola And Nigeria'. *The Extractive Industries and Society* 1 (2), 137-146.

Petroleum Industry Fiscal Bill (2018). 7, 14-15,75-77.

Rosenblum, D. (2011). Analyzing Angola's Economy: Recent past, present and future. [On-line] Available: http://www.consultancyafrica.com/index.php?option=com_content&view=article &id=692:analyzing-angolas-economy-recent-past-present-and-future- &catid=87:african-finance-a- economy&Itemid=294. [26 June 2015].

Rutledge, I. and P. Wright (1998). "Profitability and taxation: Analyzing the distribution of rewards between company and country." Energy Policy. 26(10): 795-812.

Sani Saidu and Abdul Rasheed Mohammed (2014). *The Nigerian Petroleum Bill*: *An evaluation of the effect of the proposed fiscal terms on investment in the upstream sector*. Journal of Business and Management sciences. Vol 2, No 2, 45-57.

Sonangol, (2015). *Sonangol EP-About Sonangol EP* [online] available from <http://www.sonangol.co.ao/English/AboutSonangolEP/Pages/About-Sonangol-EP.aspx> [27 June 2015].

Squire, L. and Tak, H.G., (1979). Economic Analysis of Project, J. Hopkins, Baltimore, 1979.

Tordo, S. (2007). *Fiscal Systems for Hydrocarbons*. Washington, D.C.: World Bank.

Weber, Jeremy G., (2012). *The effects of a natural gas boom on employment and income in Colorado, Texas, and Wyoming*. Energy Econ 34 (5), 1580–1588

Worldbank (2014). The Pulse: An analysis of issues shaping Africa's economic Future [online] available from http://www.worldbank.org/content/dam/Worldbank/-document/Africa/Report/Africas-Pulse-brochure_Vol9.pdf [29 May 2015].

Worldfinance, (2015). *DETAIL: Nigeria's Oil and Gas Sector Attracts FDI* [online] available from <http://www.worldfinance.com/strategy/detail-nigerias-oil- and-gas-sector-attracts-fdi> [29 May 2015].

Zahidi, S. (2010). *'Comparative Analysis of Upstream Petroleum Fiscal Systems Of Pakistan, Thailand And Other Countries With Medium Ranked Oil Reserves'*. Proceedings of the International Conference on Energy and Sustainable Development: Issues and Strategies (ESD 2010).